

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
8	(a)	(i)		it is acting as a base / it removes protons			1	1		
		(ii)		nucleophilic substitution	1			1		
		(iii)	I	M_r of —CHO fragment is 29 $\Rightarrow M_r$ of R is $72 - 29 = 43$ (1) R is C₃H₇ but branched $\Rightarrow$ compound S is H₃C H H₃C —C— H C=O (1)		1		2		
			II	orange / red	1			1		1
			III	mixture melts at a lower temperature and over a greater range			1	1		1
	(b)	(i)		use a separating / dropping funnel and run off the lower aqueous layer	1			1		1
		(ii)		NaOH / I₂ or KI / NaOCl (1) yellow solid (1)	2			2		2

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		(iii)		award (1) for each correct column <table><tr><td>Protons</td><td>Splitting pattern</td><td>Relative peak area</td></tr><tr><td>CH₃CO</td><td>singlet</td><td>3 (or 1)</td></tr><tr><td>(CH₃)₃C</td><td>singlet</td><td>9 (or 3)</td></tr></table>	Protons	Splitting pattern	Relative peak area	CH ₃ CO	singlet	3 (or 1)	(CH ₃) ₃ C	singlet	9 (or 3)		2		2		
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CH ₃ CO	singlet	3 (or 1)																	
(CH ₃) ₃ C	singlet	9 (or 3)																	
		(iv)	I	the alcohol has hydrogen bonding (the others do not) (1) therefore more energy is needed to separate the molecules (hence higher boiling temperature) (1)		2		2											
			II	similar – they each show four separate signals (four environments) (1) different – the (four) signals will be in different positions (1) accept any other sensible answers		2		2											
				Question 8 total	5	7	3	15	0	5									